

Multi-Die Design Start Guide

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Overview

Any engineer on a semiconductor design project has read many articles about the power, performance, and area (PPA), functional scalability, and time-to-market advantages of multi-die designs using 2.5D and 3D technologies. The advantages are the main reason the adoption of this approach is on the rise. However, the reality is that moving to multi-die design affects many parts of the design process and requires optimized IP, tools, and methodologies to be successful on the initial project. This white paper is a practical guide for achieving first-time success. It briefly covers the advantages of multi-die design and some of the motivating factors that may tip the balance toward committing to this approach and then presents guidelines on getting started with items that must be addressed up front on the new project.

A Primer on Multi-Die Design

Any semiconductor device in which multiple independent dies are contained within a single package is a multi-die design. This approach is a natural evolution from a printed circuit board (PCB) or a multi-chip module (MCM) containing multiple devices. The rapid adoption of multi-die design, which is not a new concept, is driven by the demands of advanced applications, such as high-performance computing and AI, and two key trends in the chip industry.

The first trend is aggregation, in which a set of discrete chips sharing a PCB are integrated into a single package. New products can be assembled quickly from existing dies, especially variants targeted for specific applications, enabling a flexible solutions portfolio. Multi-die designs shorten time to market (TTM) while reducing project risk by reusing proven dies whenever possible. In addition, die-to-die connectivity within the package provides better throughput than chip-to-chip connectivity over the PCB. Aggregation increases functional components per unit area and reduces system power consumption by minimizing PCB signals. However, aggregation necessitates advanced thermal management due to increased power density. A common example is stacking memory modules on top of a processor die, enhancing memory capacity and bandwidth while reducing system-level power consumption. In a 3D stacked die integration, much more functionality is possible per unit area and die-to-die connectivity is possible at high-bandwidth and low-latency.

The second trend driving adoption of multi-die design is disaggregation of one large monolithic system-on-chip (SoC) into smaller dies, also known as chiplets. The yield would be better for multiple smaller dies than for a large monolithic die pushing the reticle limits so disaggregation might be more economical. Multi-die designs make it possible to integrate dies on different process nodes, from different vendors, or even using very different technologies such as gallium arsenide (GaAs) to be integrated in a single package. Designers can scale the digital portion of the chip to the latest node while keeping analog portions in the existing technology to maintain better performance, avoid redesign, and reduce risk. Examples of SoCs that might benefit from disaggregation include high performance computing such as central processing units (CPUs) for data centers, graphics processing units (GPUs), Internet of Things (IoT) devices (processors for mobile devices), and AI accelerators.

Figure 1 shows examples of the different types of multi-die designs available today. Foundries offer two types of solutions—traditional 2D packaging with standard bumps and advanced packaging with micro-bumps and hybrid-bumps—for 2.5D/3D options. With advanced technologies such as 3D stacking, through-silicon vias (TSVs), interposers, and hybrid-bumps, a million times more density than a 2D designs is possible.

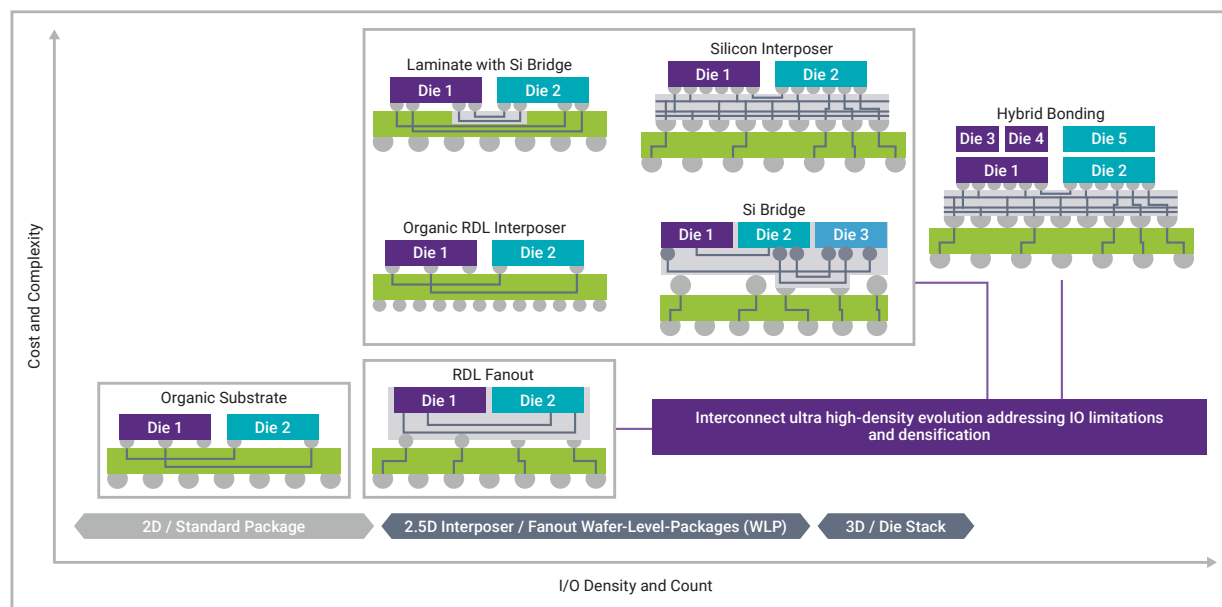


Figure 1: Types of multi-die designs and achievable I/O density

Guidelines for Project Management

Compared to monolithic designs, multi-die design complexities such as modeling, floorplanning, verification, and debugging are exacerbated. Therefore, the design team must plan to define the functional requirements very early in the design process. There may be multiple options to move from 2D to 2.5D/3D partitioning, making early feasibility studies essential. Early architecture exploration is crucial for selecting the best partitioning to meet PPA requirements, prevent thermal issues, enable die testing, ensure successful package assembly and testing, and improve package yield estimation efficiency. As the team signs off on the chosen platform, they can begin to make choices on die stacking, bump planning, TSV planning, and interposer design.

Much of this work can occur in parallel with the design of the individual dies, each of which follows traditional 2D methodologies. These include register transfer level (RTL) design and verification, software development, IP integration, hardware-software verification, implementation, signal analysis, and power estimation. However, new individual die designs cannot be completed and signed off safely without performing multi-die verification and analysis of the whole system. Some of the new capabilities needed include die integration, multi-die thermal analysis, die-to-die routing, signal and power integrity analysis, power calculations, multi-die electromigration and IR drop analysis, and a multi-die test methodology.

Selecting the right ecosystem partner is critical for a successful multi-die design, beginning with a trusted silicon foundry offering with proven 2.5D and 3D advanced packaging options and test vehicles. Since multi-die design is a new domain for most projects, it is important to choose the right partners. This starts with the silicon foundry, which must have a proven test chip using 2.5D and 3D stacking technology options. For outsourced semiconductor assembly and test (OSAT), available package types, costs, established multi-die testing methodologies, capacity, and team capabilities must all be considered when selecting the vendor. In some cases, it may be necessary to partner with a memory supplier to develop an effective High Bandwidth Memory (HBM) design strategy or help to design memory chiplets. Any IP vendor must provide silicon-proven IPs with robust test methodologies, accompanied by design and verification collateral to ensure proper integration. The chosen electronic design automation (EDA) partner must have proven and certified end-to-end reference flows with customer proof points. They must offer best-in-class applications engineering, design services options, and R&D support as needed. Leading EDA vendors can also provide silicon-proven IPs, such as Universal Chiplet Interconnect Express (UCIe), HBM, and other 3D-enabled interface IP.

Guidelines for Architecture Design

The most fundamental decision in multi-die design is how the design will be optimally partitioned in the early architecture phase. The ideal partitioning depends on the application, but the most popular strategy is based on functionality. For example, a die might contain a processor core, I/O functions, memory, or an AI engine. It is important to experiment with different partitions early in the development process to choose the one that best achieves the target PPA. In some cases, multiple partitions might be selected to produce related products for different end applications. One of the advantages of multi-die design is the ability to assemble new designs in new packages quickly without the long fabrication time required for a monolithic SoC. Generally, the architect must also select the process node for each chiplet based on PPA requirements while taking into account foundry capacity. Another important consideration is the compatibility of the chosen process node with the overall system design and the ability to meet performance targets effectively.

The architecture design phase drives several key decisions in addition to the partitioning and packaging. The memory hierarchy and coherency scheme are selected at this early stage. Although off-the-shelf chiplets are only now becoming available, in the future selecting these will also be important. The architects must choose between silicon and organic interposers and ensure that testing requirements are met. This means that the design will be testable during and after assembly, including monitoring in the field using Silicon Lifecycle Management (SLM) technology. Finally, the design must conform to system-level requirements for power, cooling, IR drop, electromigration, and mechanical stress. Figure 2 shows the range of packaging options that must be supported by any multi-die solution.

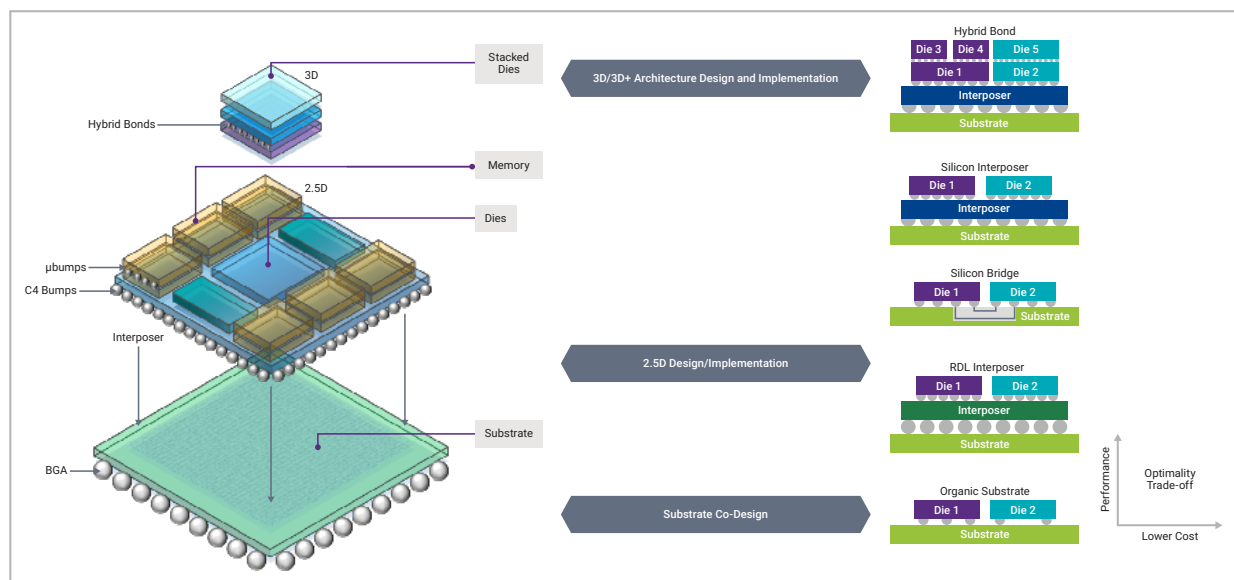


Figure 2: Packages enabled by a complete multi-die solution

The solution selected for the early architecture phase must provide the ability to try multiple partition configurations quickly and easily. Important assessments during architectural partitioning include performance requirements, power constraints, area utilization, and overall system scalability. These can be assessed with analysis, simulation, benchmarking, and feasibility studies, ensuring that the chosen partitioning aligns with project goals and technical requirements since they directly impact the design's functionality. Architects in collaboration with package designers start with simple die dimensions and package parameters for feasibility studies and refine them as they make key decisions on partitioning and packaging. The platform should include pre-existing templates for IP models, with die models being added as die-level IP becomes available. The solution must enable the creation of virtual prototypes for each partitioning. This allows pre-production software development to begin using an emulation system or an FPGA prototyping solution. Since software development is often a gating item for final product release, starting it long before chips are available in the lab accelerates time to market.

Guidelines for Multi-Die Verification

Verification of a multi-die design requires a superset of the capabilities needed for a single SoC. The verification strategy must still encompass the IP, subsystem, and die levels, not necessarily using the same tools and methodologies as traditional monolithic designs. However, verification must extend to the full multi-die design to verify die-to-die connectivity and cross-die functionality. Key aspects in multi-die verification include validating architecture assumptions, die-to-die connectivity, and interface verification. Additionally, monitoring, integration, testing, repair, and interface IP integration are crucial. These tasks can be performed as part of the architecture phase, including verification based on workloads such as AI, networking, I/O, and memory. All system tests must be validated using the full multi-die design. The ability to export to emulation and virtual prototyping is critical to handle the large capacity required for a full system, provide a reasonable execution speed, and verify the hardware and software together. The selected EDA and IP vendor partner for verification solutions must provide:

- Relevant verification IPs
- A multi-die platform including the verification IPs
- Workload-based testing
- System architecture testing
- Testing based on interface IP protocols: reset, boot, debug, power management, security, reliability and serviceability (RAS), etc.

Guidelines for Implementation, Analysis, and Signoff

Although implementation of each die follows the same process as traditional SoC designs, the design implementation phase also applies to the complete multi-die design. This includes die-to-die routing and packaging design, based on the results from the early architecture exploration stage. Timing, power, clock signals, and thermal characteristics need to be assessed concurrently during the design process. This is achieved by using optimized tools and methodologies that allow for the simultaneous analysis to be executed in the design environment.

It must be possible to continuously analyze PPA and thermal characteristics in different design phases to ensure compliance with end-product requirements. The chosen unified exploration-to-signoff platform must support hierarchical place and route, die-to-die interface (e.g., HBM, UCIe) auto-routing and shielding, manual and auto-routing redistribution layer (RDL), power-ground (PG) mesh routing, and metal-insulator-metal (MIM) capacitor insertion with PG alignment.

The final stage of multi-die design is signoff, which covers a wide range of analysis at the full-package level, including the steps shown in Table 1. The white paper "[Achieving Successful Timing, Power, and Physical Signoff for Multi-Die Designs](#)" provides a detailed explanation on how to address timing, power, and physical signoff requirements for multi-die designs.

Implementation	Signoff
• Packaging consideration – Silicon/RDL-based interposer (2.5D) – Direct die stacking (3D) – FOWLP/organic substrate • Multi-die prototyping and floorplanning – Multi-die top-level netlist creation – Die placement – Die orientation • Bump planning and mirroring • TSV planning • Chiplet and interposer/substrate/package power network design • Die-to-die routing – RDL – HBM, UCIe and 3DIO	• Design rule checking (DRC) • Layout versus schematic (LVS) checking • Extraction • Static timing analysis • Engineering change orders (ECOs) • Power checking • Thermal checking • Power Integrity analysis • Signal Integrity analysis • Thermal Integrity analysis • Electromigration analysis • IR drop analysis

Table 1: Analysis steps for multi-die designs

Guidelines for Required Tool Capabilities

Moving to multi-die design impacts many of the EDA tools in the development flow. The previous sections itemized some of the capabilities required. It is important to have a holistic view of the multi-die die/package co-design and co-optimization environment, using a single platform from a single vendor whenever possible, as shown in Figure 3. This platform must include solutions for:

- Early architecture exploration for feasibility and prototyping
- RTL and die implementation
- Multi-die verification
- Interposer, die stacking, bump planning, TSV planning, and 3D checking
- Timing and power signoff
- Physical verification with system-level analysis
- Virtual prototyping
- Manufacturing test
- In-field SLM monitoring



Figure 3: Multi-die design methodology from exploration to signoff

Beyond EDA tools, the selected partner must provide other capabilities essential for successful multi-die design. High-bandwidth UCIe IP proven on test chips provides fast, reliable die-to-die connectivity. High-quality HBM IP meets the essential high-bandwidth and low-power memory requirements of multi-die designs. SLM IP, supporting the IEEE 1838 test access infrastructure for intra-die lane testing, enables manufacturing testability, yield improvement, in-field testing and optimization, in-field monitoring for aging, and preventive maintenance. Packaging experts can provide advice and help with the decisions on interposers, OSAT selection, and more. Applications engineers, backed by the services and R&D teams, provide all the training and support needed for project success.

Summary

Multi-die designs are becoming increasingly common, driven by both disaggregation of large SoCs and aggregation of multiple discrete chips into a single package. This provides many advantages in terms of greater design flexibility, functional scalability, increased density, lower system power, and higher performance. However, multi-die design presents some challenges beyond traditional monolithic SoCs. This white paper has summarized some guidelines that must be followed to handle these challenges and increase the chances of first-time success with this relatively new approach. It has also provided advice in choosing partners, including an EDA and IP vendor with real-world experience and a strong track record in multi-die design.

As a trusted partner, Synopsys is driving the industry transformation to multi-die designs with a comprehensive and scalable solution for fast heterogeneous integration. The Synopsys solution, including EDA and IP products, enables early architecture exploration, rapid software development and system validation, efficient die/package co-design and optimization, robust die-to-die and chip-to-chip connectivity, and improved manufacturing and reliability. This solution meets all the requirements outlined in the previous sections of this white paper.

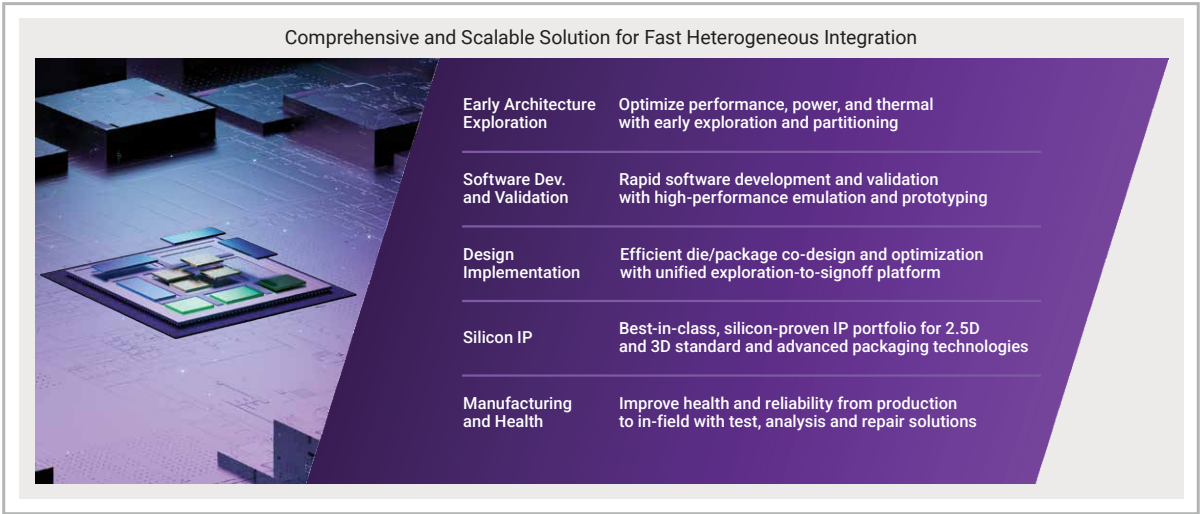


Figure 4: Synopsys multi-die solution for fast heterogeneous integration

Any project team considering or planning to adopt multi-design must clearly understand the benefits and challenges, follow the guidelines in this white paper, and partner with Synopsys for the easiest path and the best outcome.